

Priced In, Bit by Bit: Measuring Information Aggregation in Prediction Markets

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Prediction markets are typically evaluated on whether their prices are well calibrated forecasts of eventual outcomes. This paper asks a different question: how informative are prices? Using Shannon entropy and mutual information on a panel of roughly 50,000 closed binary markets on Polymarket, we find that observing prices on average resolves 38.3% of uncertainty surrounding outcomes, and less in markets that are part of a set of mutually exclusive outcomes. Among markets open for over four months, which are more informative than average, prices reduce 42.2% of uncertainty four months from resolution, rising to 85.4% one day from resolution. Finally, following Samuelson’s martingales, we show that while price changes are statistically predictable, any departure from a fair game is economically negligible. These findings are largely consistent with Hayek’s claim that prices aggregate dispersed information.

Keywords: prediction markets, information aggregation, market efficiency, mutual information, entropy

JEL Classification: D83, G14, C13

1. Introduction

Despite their recent surge in popularity, prediction markets are not a novel idea. The earliest recorded example of a ‘prediction’ market goes all the way back to 1503, where people bet on the papal elections ([Rhode and Strumpf, 2013](#)). Five centuries later, traders on Polymarket wagered millions on who would ascend to the Papacy. Since ([Hayek, 1945](#)), economists have recognized that prices serve dual roles: they both allocate resources and aggregate information on the value of those resources. Prediction markets aim to exploit the information aggregating power of prices in an attempt to forecast future events, as argued by a wide range of leading economists ([Arrow et al., 2008](#)).

A price $p \in [0, 1]$ in a prediction market tells us how likely the market believes an outcome Y is. We’ll restrict ourselves to binary market outcomes only, so $Y \in \{0, 1\}$ (either the outcome doesn’t happen, or it does), with $Y = 1$ priced at p and therefore $Y = 0$ priced at $1 - p$. The equilibrium price p equals (or is very close to) the mean beliefs of the traders that the event will occur ([Wolfers and Zitzewitz, 2006](#)). The first modern academic prediction market was the University of Iowa’s Iowa Electronic Markets, introduced during the 1988 US presidential election. Although prediction markets contain many irrational or mistake-prone traders, empirical evidence from the Iowa Electronic Markets suggests that market prices remain highly informative because marginal traders responsible for price formation are substantially more rational than the average participant and arbitrage disciplines persistent pricing errors ([Oliven and Rietz, 2004](#)).

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Prediction markets have a distinctive property over other financial markets: they are one of the very few financial environments in which market efficiency can be evaluated directly because the hidden state of the world is eventually revealed (Thaler and Ziemba, 1988). Consider a stock X . Is the stock correctly priced today? There's no objective answer to that question. The stock's fundamental value is a theoretical object that remains unobservable forever. On the other hand, consider the event $Y = \text{"party } A \text{ wins the election."}$ Y starts out unknown, but eventually, nature will reveal $Y \in \{0, 1\}$ and thus, Y will be observed.

The fact that prediction market prices can be tested against an observed realization of outcomes fundamentally changes what can be tested empirically. In conventional financial markets, the Efficient Market Hypothesis (Fama, 1970) is typically evaluated only indirectly through implications such as return predictability, abnormal returns, or event studies, since the underlying fundamental value is never observed. Prediction markets instead allow direct evaluation of informational efficiency because realized outcomes become available after market resolution.

Existing evaluations of prediction markets have focused primarily on calibration, scoring rules, and forecast accuracy. While these establish whether prices are statistically well calibrated, they do not quantify how much uncertainty market prices eliminate about the eventual outcome. This paper adopts an information-theoretic perspective, measuring the reduction in uncertainty about the eventual outcome attributable to market prices through Shannon entropy and mutual information (Shannon, 1948). Hayek's central claim was that prices aggregate dispersed information. If prices aggregate information, then observing prices should progressively make us less uncertain about the future state of the world as we accumulate new information. Shannon defined information itself as a reduction in uncertainty produced by an observation, making Shannon entropy a natural metric to measure how much information prediction market prices contain.

This paper answers three questions regarding the informativeness of prices. Are they informative at all? If yes, how does the informativeness of prices change over time, and finally, how do prices become informative in the first place? First, we quantify how much uncertainty Polymarket prices resolve on average using mutual information. Second, we trace how this informativeness accumulates over time by measuring the mutual information across various time to resolution bins. Third, we use (Samuelson, 1965)'s martingales to test whether price changes can be predicted, which would negate efficient incorporation of information.

We find that prices resolve roughly 53.65% of outcome uncertainty on average, with this figure remaining consistent across Negative Risk markets too. Among markets that've been open for > 4 months, markets resolve 36.36% of outcome uncertainty even > 4 months from resolution, while this figure increases as time to resolution falls. We also see that while price changes can be predicted with statistical significance (as a result of the large sample size), any deviations from prices being a martingale are minuscule at best and cannot reliably be exploited for profit.

2. Methodology

2.1. Why information theory?

If we'd like to quantitatively study evidence for the Hayekian claim that prices aggregate dispersed information, a field named "Information Theory" is hardly a surprising choice. Hayek's

theory is fundamentally about communicating information. If prices can in fact communicate information, it should leave us better informed about the eventual realization of an event (which prediction markets have the unique feature of eventually observing). In simple terms, prices make us better informed, i.e. less uncertain about what will happen. Shannon's theory provides a natural framework because it gives us a way to quantitatively measure information as the *uncertainty reduced* regarding an outcome. This raises the empirical question of how we can quantify uncertainty and its reduction.

(Shannon, 1948) proved that if we have a set of possible events whose probabilities of occurrence are $p_1, \dots, p_n$, with the probabilities being the only thing we know about which event will occur, then the only measure of how much choice is involved in the selection of the event, i.e. how uncertain we are of the outcome, that satisfies a few mathematically desirable properties is of the form

$$H = -K \sum_{i=1}^n p_i \log p_i$$

with $K = 1$ being defined as the *entropy* of the set of probabilities $p_1, \dots, p_n$. Taking $\log_2$ gives entropy measure a unit he called "bits". Entropy is highest when both outcomes are equally likely, since the outcome is most difficult to predict, and falls as one outcome becomes increasingly certain. Thus, it provides an intuitive quantitative measure of uncertainty. Since Polymarket prices are just probabilities (in the binary case, just p_1 and p_2), we can define the entropy of an outcome Y on Polymarket to be $H(Y)$.

Economic logic tells us market prices provide additional information beyond the unconditional probability of an event. After observing the market price, uncertainty about the eventual outcome should decline if prices aggregate information successfully. We denote this as the *conditional entropy* of event Y given the market price using the notation $H(Y|P)$. Mathematically, $H(Y|P)$ is the average entropy of Y for each value of P , weighted by the probability of getting that particular value of P :

$$H(Y|P) = - \sum_p \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, P = p) \log_2 \mathbb{P}(Y = y|P = p)$$

Now that we have a measure for the unconditional uncertainty of Y and the conditional uncertainty upon observing the price, we can finally mathematically define information as uncertainty reduction. Shannon claimed "the rate of actual transmission [of information], R , would be obtained by subtracting from the rate of production (i.e., the entropy of the source) the average rate of conditional entropy." Since prices act as signals in prediction markets, the information transmitted by it is given by

$$I(Y; P) = H(Y) - H(Y|P)$$

Note that what Shannon defined as R , we'll stick to the modern convention and call this quantity *mutual information*, or just information for convenience. Mutual information will be our main object of interest in this paper. A large value tells us that observing prices dropped uncertainty by a large amount relative to the uncertainty of the unconditional distribution of outcomes, thus, larger values of mutual information are more desirable.

2.2. Information vs calibration

There have been numerous studies that have concluded that besides a few behavioral biases (like the favorite longshot bias), prediction markets are generally accurate and well calibrated, often outperforming professional forecasters and polls in a variety of statistical tests. Studies that show this have been summarized in (Snowberg et al., 2012). Our objective, however, is different. Rather than asking whether prices are *accurate* forecasts, we ask how *informative* they are. Consumers of economic forecasts care not just about the accuracy of the mean prediction, but the uncertainty attached to it as well. The standard proxy for uncertainty is disagreement among forecasters. As (Snowberg et al., 2012) show, that is a poor proxy for actual uncertainty. Using information theory, we provide an alternative way to measure uncertainty in markets.

The difference between information and calibration is one that has real implications; it isn't just a semantic difference. Consider a market for an election between two equally likely candidates A and B that prices candidate A at 0.5. Observing the price hasn't given us any new information, as the unconditional distribution of outcomes is roughly 50% anyways. Suppose a major news story breaks out and pushes candidate A 's price to 0.95. This gives us a lot of new information, as we now know that the market thinks A is almost certain to win. Notice how the informativeness of a market can be established before any calibration metric can be analyzed. Calibration evaluates the quality of a prediction *after* the uncertainty has been resolved. Information measures how much of that uncertainty was eliminated *while* the outcome was still unknown. These are complementary, but fundamentally different, dimensions of market quality.

3. Data

3.1. Source and Construction

Our dataset combines information from Polymarket's Gamma API, which gives us cross-sectional market data, and the Polymarket CLOB (Central Limit Order Book), which gives us hourly snapshots of the price history across our markets.

To calculate information-theoretic quantities, we need a well defined random variable whose realizations are eventually observed. This restricts our sample to closed markets only. We further restrict our sample to closed binary markets. This keeps the interpretation simple as for an outcome Y , $p = P(Y = 1)$ and $1 - p = P(Y = 0)$. Interpretation gets more complex as we move on to multi-outcome markets and keeps our analysis a lot cleaner. As these markets are already resolved, there is no trading activity on them and hence the Gamma API reports a liquidity of zero on almost all these markets. Hence, we do not use liquidity in our analysis.

The initial Gamma dataset contained 603,409 binary resolved markets, with data ranging from the start of Polymarket's Gamma data availability till July 18th, 2026. We then sort the data by volume and only select the top 50,000 markets by volume (roughly 71.5th percentile) to be part of our sample. This ensures that our analysis is not skewed by uninformative low volume markets. Although we started out with 50,000 observations, we had to drop 14 observations with no volume, 3 observations without a specified market close time and a further 12 observations that did not have a price of 0 or 1, i.e. did not cleanly resolve, leaving us with 49,971 markets in our sample. The market IDs of the sample are then matched to the price history data from CLOB to get a panel with historical price data for each market in our sample. This process leaves

us with a panel that is indexed by market ID and timestamp (hourly) and has 43.4 million rows observations corresponding to 49,971 markets.

For each observation, the remaining time until market resolution is computed as the difference between the observation timestamp and the market closing time. This variable serves as the primary temporal dimension of the analysis. To facilitate comparisons across markets with widely differing lifetimes, time to resolution is sorted into equal length bins. Prices are also binned into equally spaced 5 percentage point intervals to estimate how information measures vary across different predicted outcome probabilities.

3.2. Summary Statistics

Table 1: Market Summary Statistics

Variable	Mean	Median	SD	Min	Max
Trading volume (\$ million)	1.126	0.160	10.984	0.052	1531.479
Market duration (days)	38.727	13.385	64.309	0.088	939.485
Resolved Yes (=1)	0.286	N/A	N/A	N/A	N/A
Negative Risk (=1)	0.624	N/A	N/A	N/A	N/A

Table 2: Panel Summary Statistics

Variable	Mean	Median	SD	Min	Max
Price	0.203	0.070	0.270	0	1
Time to resolution (days)	18.77	6.522	31.034	0.007	294.957

4. Question 1: Are prices informative?

The methodology section has given us a quick primer on a quantitative definition of information. Now it is time to apply that and answer the central empirical question of this paper: how informative are these prediction markets?

4.1. Estimation

Mutual information is a property of a joint distribution of outcomes and prices, so before estimating it we must say which distribution we mean. Our panel contains one observation per market per hour, and markets differ enormously in how long they trade: the median market lasts 13 days, while some last more than two years. If every market-hour counted equally, long-lived markets would dominate the estimate. We instead let every market count equally. Market i is observed for T_i hours, and each of its hours receives weight $1/T_i$, so that each market's weights sum to one. With N markets, the joint distribution of outcomes and binned prices is

$$\hat{\mathbb{P}}(Y = y, P \in b) = \frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t=1}^{T_i} \mathbf{1}[Y_i = y, p_{it} \in b],$$

and $H(Y)$, $H(Y|P)$ and $I(Y;P)$ are all computed from this one distribution. The resulting quantity can be interpreted as the information contained in the price of a typical market, averaged over that market's trading life.

Price is continuous, so we discretize it into 20 bins of width 0.05 and compute $H(Y|P)$ as the entropy of outcomes within each price bin, averaged across bins using the weights above. Binning can only discard information, so in the population $I(Y; \text{bin}(P)) \leq I(Y; P)$ and our estimates are conservative. In finite samples, plug-in entropy estimates are biased downward, which biases mutual information upward. We apply the [Miller \(1955\)](#) correction, which adds $(K - 1) / (2N \ln 2)$ bits to an entropy estimated from N observations of a variable taking K values. Because every hour of a market shares the same outcome, the independent units are markets rather than market-hours, so N is the number of markets. The correction lowers $I(Y; P)$ by about 0.0003 bits in the full sample.

4.2. Results

Table 3 reports the results. A binary outcome has entropy $H(Y) = -(q \log_2 q + (1 - q) \log_2 (1 - q))$, where $q = \mathbb{P}(Y = 1)$, which reaches its maximum of 1 bit at $q = \frac{1}{2}$. In our sample 28.6% of markets resolve Yes, giving a baseline entropy of $H(Y) = 0.8635$ bits, or 86% of the maximum possible uncertainty for a binary event. Therefore, without looking at prices, outcomes are highly unpredictable.

Conditioning on the price lowers the uncertainty to $H(Y|P) = 0.5329$ bits, so prices carry $I(Y; P) = 0.3303$ bits of information about the eventual outcome. This tells us that observing the price of a typical market at some point in its life resolves 38.3% of the uncertainty about how it will resolve. Prices are clearly informative, although a majority of the uncertainty remains until the outcome is known.

Our choice of weights also matter significantly. Markets differ in their lifespan, so if we directly calculate information on our dataset, markets with longer lifespans show up more in the data, and thus our result will be skewed towards such markets. Therefore, our baseline specification weighted markets based on their lifespan, so each market is counted the same in our result. If we skip this and directly report information, we get the hour-weighted metric as markets with longer hours are weighted more. Doing so reveals prices still carry a similar amount of information, 0.31 bits compared to 0.33, but it is a larger share of the uncertainty, 43.7% compared to 38.3%. This difference is largely due to $H(Y)$ being smaller in longer lived markets, which disproportionately resolve No. Only 19.5% of market hours belong to markets that resolve Yes, despite being 28.6% of all markets. This makes our hour-weighted sample dominated by more predictable markets, thereby starting with less uncertainty.

Table 3: Information in Polymarket prices

	Markets	$\mathbb{P}(Y = 1)$	$H(Y)$	$H(Y P)$	$I(Y; P)$	% of $H(Y)$
Baseline (market-weighted)	49,661	0.286	0.8635	0.5329	0.3303	38.3
Hour-weighted	49,661	0.195	0.7108	0.4002	0.3103	43.7

Notes: Entropies and mutual information are in bits. Prices are grouped into 20 bins of width 0.05. In market-weighted rows, each hour of market i receives weight $1/T_i$, where T_i is the number of hours the market is observed, so each market counts once; in the hour-weighted row, every market-hour counts once. $\mathbb{P}(Y = 1)$ is the weighted share of Yes outcomes. $I(Y; P)$ includes the Miller–Madow correction with N equal to the number of markets, so it is about 0.0003 bits smaller than $H(Y) - H(Y|P)$.

4.3. Negative Risk Markets

Negative risk is a market structure where a combination of positions can reduce risk to zero and potentially create a guaranteed profit by exploiting mispricings. Polymarket has many such markets, usually when dealing with a set of mutually exclusive outcomes. For example, we could have three separate markets for the same award: movie $A/B/C$ wins the Oscar for Best Picture, with each of the three markets being a binary yes/no market for each movie. Assuming the Academy only awards the title to one movie, which it always has done, then one and only one of these 3 markets will resolve to a yes. If you buy a yes contract on all 3 markets, you are guaranteed a payoff of a dollar from the market that resolves yes, while gaining nothing from those that resolved no.

It is important for Polymarket to keep track of markets tied by negative risk due to the presence of arbitrage. If the cost of a yes contract on all 3 markets is less than a dollar, you’ve got a free profit by buying all three yes contracts. Polymarket has a mechanism to convert no shares into yes shares across such markets, allowing arbitrage to push yes prices towards summing up to 1. This presents a problem for our information aggregation story as now prices can change due to arbitrage, even without the introduction of new information into the market.

Table 4: Information in prices by market structure

	Markets	$\mathbb{P}(Y = 1)$	$I(Y; P)$	% of $H(Y)$ resolved		
				Baseline	Excl. final 24h	Hour-weighted
All markets	49,661	0.286	0.3303	38.3	34.5	43.7
Negative risk	30,999	0.236	0.2473	31.4	28.0	43.2
No negative risk	18,662	0.369	0.4305	45.3	41.2	43.8

Notes: The two groups split the full sample and together make up the “All markets” row. Markets, $\mathbb{P}(Y = 1)$ and $I(Y; P)$ (in bits, Miller–Madow corrected) refer to the baseline, market-weighted specification. The last three columns report the share of $H(Y)$ resolved by prices under the three specifications of Table 3. Excluding the final 24 hours removes markets that traded for less than a day (30,883 and 17,989 negative-risk and other markets remain).

Table 4 splits the sample into negative-risk and other markets. The two groups differ sharply in baseline uncertainty. Only 23.6% of negative-risk markets resolve Yes, compared with 36.9% of other markets, so negative-risk outcomes are more predictable to begin with ($H(Y) = 0.788$ against 0.950 bits). This isn’t surprising, as a set of mutually exclusive markets generally has a few genuine contenders and many longshots that are priced near zero and resolve No. As in our

Oscars example, even though very few movies are considered serious contenders for Best Picture, markets for many more movies than that may exist on Polymarket.

Prices in negative-risk markets are also less informative. They carry 0.25 bits of information about the outcome, compared with 0.43 bits in other markets, and resolve 31.4% of the uncertainty compared with 45.3%. The gap is not driven by trading after outcomes are known: excluding each market's final 24 hours gives 28.0% and 41.2%. It does, however, disappear when every market-hour counts equally, which gives 43.2% and 43.8%. Hour weighting emphasizes long-lived markets, so the difference comes from short-lived markets: among markets that trade for a long time, prices resolve about the same share of uncertainty under both structures, while short-lived negative-risk markets are markedly less informative than other short-lived markets. Whether this reflects arbitrage-driven price movements or the kinds of events that short-lived negative-risk markets cover is a question we leave for future work.

5. Question 2: The dynamics of informativeness

The previous section established that market prices eliminate over half of the uncertainty surrounding future outcomes on average. However, an average alone tells us little about when this information enters prices. It could come in months in advance, or when an event happens that makes the outcome obvious. Prediction markets telling us that an event will happen after the general consensus has ruled it as extremely likely is not very informative. An informative market must certainly have the ability to be informative long before the event is considered "settled".

5.1. Measuring information at a given horizon

We measure how much the information the price contains d days before a market closed contains, denoted by $I(Y; P_d)$ for $d = 1, 2, \dots$ days. For each market and day d , we take the last hourly price recorded at least d days before the market is closed, so each market contributes one observation at each day. Estimation follows the same process as in Section 4, using 20 price bins and Miller-Madow correction.

Before we plot information against time, we must be mindful a potential selection bias: markets have different lifespans. All closed markets must, by definition of being closed, have existed an hour before resolution, while not all markets need to be 4+ months old. Directly plotting would give us a different sample size in each time to resolution bin. We therefore work with balanced cohorts: the cohort for a lifetime of D days contains the markets that traded for at least D days, and the same markets appear at every horizon from D days to one day. We use four cohorts, with D equal to one week, one month, two months and four months. We also require every market to have traded for at least 48 hours before the cohort's earliest horizon, so that no market enters the curve on its opening day, when its price is still largely the initial quote and is largely uninformative.

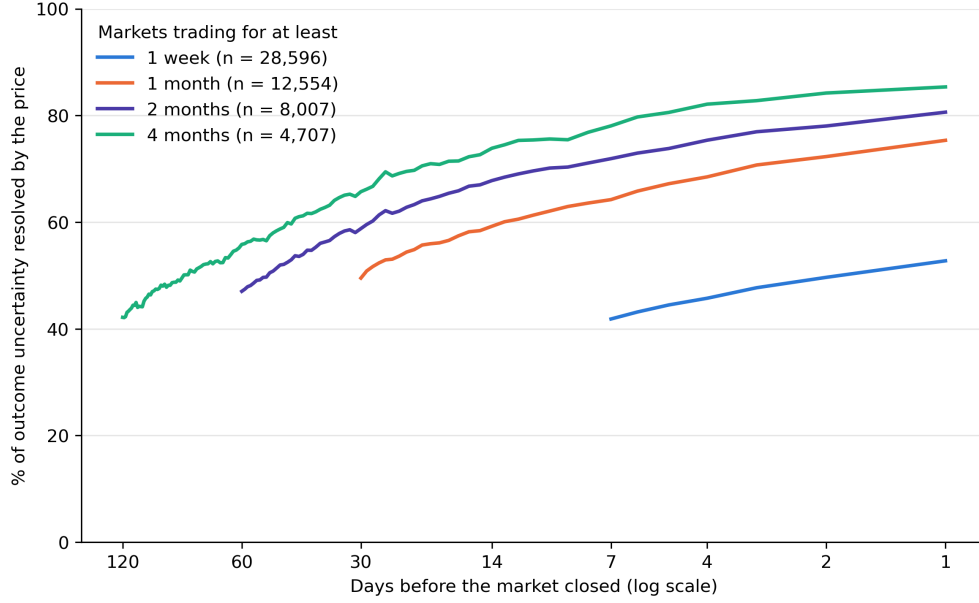


Figure 1: Share of outcome uncertainty resolved by the price, by days before the market closed.

Figure 1 plots the share of outcome uncertainty resolved by the price against the number of days before close, and Table 5 reports the four-month cohort at selected horizons. Prices in long-lived markets are informative early. Four months before closing, prices in the four-month cohort already resolve 42.2% of the uncertainty about the outcome. This share rises steadily to 65.7% one month before close, 78.1% one week before, and 85.4% one day before. Information aggregation is a continuous process that begins long before outcomes become obvious.

Table 5: Information in prices by horizon, four-month cohort

Days before close	$H(Y P_d)$	$I(Y; P_d)$	% of $H(Y)$	Gain per day
120	0.3428	0.2519	42.2	–
60	0.2610	0.3336	55.8	0.23
30	0.2019	0.3926	65.7	0.33
14	0.1533	0.4413	73.9	0.51
7	0.1283	0.4660	78.1	0.60
4	0.1038	0.4899	82.1	1.36
2	0.0914	0.5024	84.2	1.04
1	0.0849	0.5096	85.4	1.14

Notes: Markets that traded for at least four months ($n = 4,707$). $H(Y) = 0.597$ bits at every horizon (14.5% of these markets resolve Yes). $I(Y; P_d)$ includes the Miller-Madow correction. “Gain per day” is the change in the share of $H(Y)$ resolved since the previous row, in percentage points, divided by the number of days between the two horizons.

At every horizon, prices in longer lived markets resolve more of the uncertainty. A potential explanation is the kind of event each cohort covers. Shorter markets may tend to include more volatile events like close sports games, where the outcome could drastically change a few hours before resolution. On the other hand, longer lived markets deal with questions that prices identify as unlikely well in advance, as only 14.5% of the markets in the 4 month cohort resolve Yes.

5.2. Rate of information aggregation

To measure changes in information aggregation, we can plot the average information gain per day in each time period against the time periods directly. Simply using the daily change in informativeness is largely noise and is not very informative, thus we bin days into different time periods.

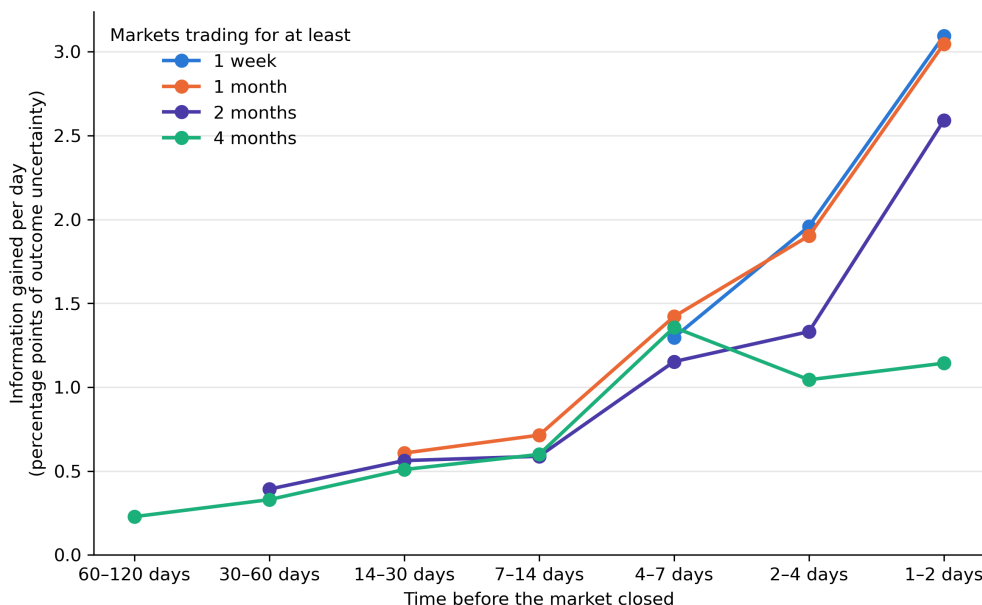


Figure 2: Average information gained per day, in percentage points of outcome uncertainty, in each stretch of time before the market closed.

Prices clearly learn faster as resolution approaches. In the four month cohort, prices resolve only 0.23 percentage points of uncertainty per day between four and two months to close, and 1.36 per day in the final week, a roughly sixfold increase. However, the gain in informativeness levels off to around 1 percentage point per day in the final week. All the other markets clearly exhibit increasing rates of gain in informativeness, suggesting that information aggregation on Polymarket speeds up over time in most markets.

These results show that information aggregation in prediction markets is on average a gradual process, not confined to the moments right before resolution. Markets do contain substantial amounts of information many months before resolution. The average informativeness reported in the previous section is therefore not driven solely by prices immediately before market resolution but reflects information that has already accumulated throughout the market's lifetime. The results also suggest that the price's stock of information is increasing at an increasing rate, although this is hard to definitively establish using just discrete approximations of change.

6. Question 3: How do prices incorporate information?

So far, we've used mutual information to show that prices contain substantive information and how that information content increases over time. However, a natural question remains: how does information enter these prices? There are numerous plausible mechanisms. Perhaps news

is immediately efficiently incorporated into prices by rational traders. Maybe information is gradually diffused into the market over time. It is also possible that traders may initially overreact to new information, leading to the price overshooting and then returning to an equilibrium. Across all three mechanisms, we could end up with very different price dynamics but similar information trajectories.

One way we can distinguish between these cases is by looking at serial correlation in price changes. Following (Fama, 1970), if prices immediately react to and incorporate new information, then you wouldn't expect any persistent pattern in price changes. Changes in price would show zero serial covariance (and thus no autocorrelation). On the other hand, gradual diffusion of positive news would lead to a persistent upward pattern in price changes, and overreaction would lead to a correction that shows persistent downward price changes, and vice versa for negative news.

For lags $k \in \{1, \dots, 24\}$, we compute the correlation between the current price change Δp_t and the k -hour lagged price change Δp_{t-k} per market. This gives us the following plot for autocorrelation:

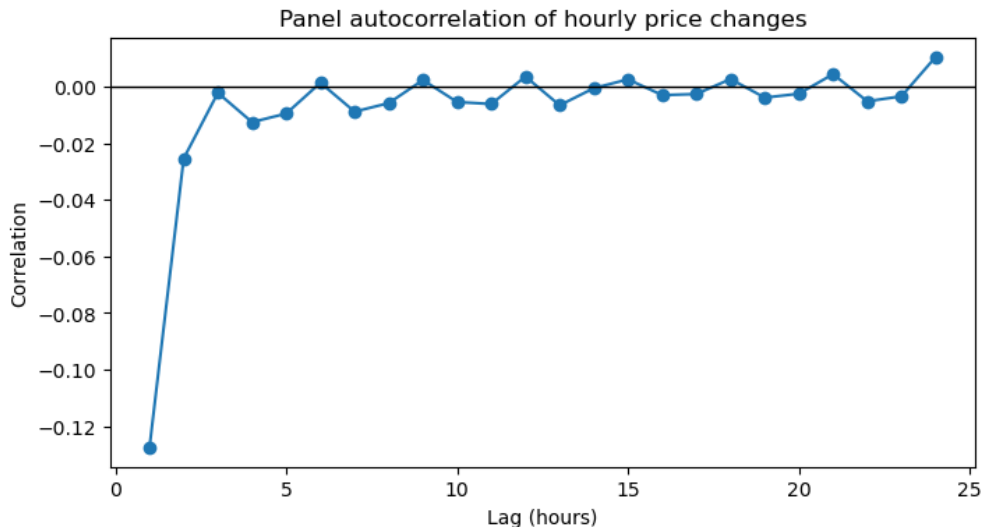


Figure 3: Panel autocorrelation of hourly price changes.

We observe slight negative autocorrelation at the one hour gap, after which any autocorrelation, although statistically significant, is economically negligible and almost certainly not exploitable for profit. The initial negative autocorrelation could be due to market microstructure effects (like bid-ask bounce, although Polymarket's documentation doesn't make it clear whether prices are last trade price or the bid-ask midpoint). Inefficiencies as the cause cannot be entirely ruled out, however, they are corrected in about an hour, so any possible inefficiency fails to persist.

It could be argued that our autocorrelation plot is an artifact of aggregation, where we genuinely have different types of markets that incorporate information differently, yet aggregating the results smoothens differences and presents it as looking like an informationally efficient market. To test this, we plot the coefficients β_1, β_2 and β_3 obtained from regressing Δp_t on $\Delta p_{t-1}, \Delta p_{t-2}$ and Δp_{t-3} for each market and plot the distribution of β s obtained. We see no signs

of multimodal distributions here. This tells us that autocorrelation patterns observed are pervasive and consistent across markets, rather than an artifact of pooling heterogeneous markets. Combined with the very small autocorrelation observed beyond one hour, this points towards markets efficiently incorporating information.

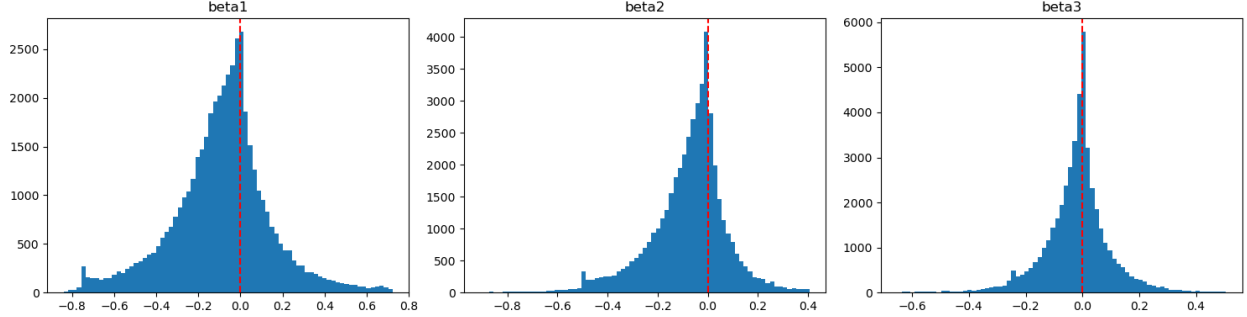


Figure 4: Distribution of autocorrelation coefficients.

6.1. Polymarket prices as a fair game

So far, we’ve tested whether a market’s own price changes can predict future price changes, i.e. Fama’s weak form efficiency. A broader result is (Samuelson, 1965), who showed that if prices are formed as the conditional expectation of an outcome given *all available information*, $p_t = \mathbb{E}[Y|I_t]$, then price changes behave like a fair game wrt the information set, i.e. $\mathbb{E}[\Delta p_t | I_t] = 0$: given everything we know, the expected future price equals today’s price. Prices can only change if new information arrives, and by definition that cannot be predicted. This asserts that no piece of information can predict future price changes.

Unfortunately, we do not have access to every single possible datapoint players in the market may use to make their decisions. Our best option is to use a wider set of variables that are known when the price change begins: the previous price, the time remaining until the market’s scheduled end date, recent volatility, and the three most recent price changes. We estimate

$$\begin{aligned} \Delta p_t = & \alpha + \gamma p_{t-1} + \delta \log(1 + \max(\tau_t, 0)) + \kappa \mathbf{1}[\tau_t < 0] \\ & + \sum_{k=1}^3 \beta_k \Delta p_{t-k} + \lambda \sigma_{t-1} + \varepsilon_t, \end{aligned} \tag{1}$$

where Δp_t is the hourly price change in cents, τ_t is the number of hours until the market’s scheduled end date, and σ_{t-1} is the standard deviation of hourly price changes over the previous 24 hours. Under the martingale hypothesis, every coefficient in (1) is zero.

Table 6: Predictability of hourly price changes

	Coefficient	Std. error
p_{t-1} (per 10 pp)	-0.0046	0.0001
$\log(1 + \tau_t)$	-0.0033	0.0002
$1[\tau_t < 0]$	0.0177	0.0048
Δp_{t-1}	-0.1429	0.0021
Δp_{t-2}	-0.0505	0.0013
Δp_{t-3}	-0.0146	0.0013
σ_{t-1} (cents)	-0.0163	0.0006
Intercept	0.0374	0.0018

Notes: Pooled OLS estimates of equation (1); the dependent variable is the hourly price change in cents. 42,353,513 observations from 48,974 markets; $R^2 = 0.021$; joint Wald $F = 977.161$. Standard errors clustered by market. All coefficients are significant at the 1% level.

43 million observations effectively makes any result statistically significant. What matters here is economic significance. Notice that even when conditioning on a much wider information set, our model can only explain 2.1% of variation in hourly prices ($R^2 = 0.021$), with every variable having very small coefficients. This tells us that while we can predict price changes with statistical significance, in practice, the magnitude of that predictability is negligible at best. The estimated effects are small enough that they’re unlikely to support profitable strategies once transaction costs and other frictions are taken into account. Consistent with Samuelson’s fair game hypothesis, publicly available past information provides little practical ability to forecast subsequent price movements on Polymarket.

7. Conclusion

This paper set out to answer three questions: whether prediction market prices are informative, how that informativeness evolves over time, and whether the process by which it accumulates looks efficient. The answers to these three questions build up on each other and form a clear narrative. Prices resolve roughly half of outcome uncertainty on average, with informativeness climbing from resolving around a third of the total uncertainty at long horizons to nearly all of it in the final day before resolution. The martingale results speak to what that remaining, unresolved uncertainty is likely *not*: our tests suggest that whatever information is already reflected in recent prices, volatility, and time to resolution is being incorporated efficiently, since none of it meaningfully predicts future price movements. What we cannot say is whether the residual uncertainty at any given point is genuinely undetermined, or whether it reflects dispersed information that simply hasn’t reached the market yet, as price unpredictability is consistent with both. We also have not yet formally tested how sensitive our mutual information estimates are to the choice of price bin width; both are natural next steps.

In general, this paper’s contribution is less about Polymarket specifically and more about a method. In markets where a latent variable that drives pricing is eventually observable (the outcome happening or not here), entropy and mutual information offer a way to ask how much

a price tells you, independent of whether that price turns out to be 'right'. As prediction markets are increasingly proposed as forecasting tools for policy and institutional decision making, we believe the distinction between a market being accurate and a market being informative is one worth taking seriously. For policymakers and other users of forecasts, this distinction matters because a market can provide useful information long before its forecast can be judged correct. Decisions have to be made before outcomes are known, so what a market tells us today can matter even when its final forecast cannot yet be evaluated.

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